

Assignment 15_Report_Loan Default Prediction

Link to Colab

<https://colab.research.google.com/drive/1cZ5BCuPFx2nRDPw-9EUg53gUv79Xkn7t?usp=sharing>

TASK 1: Data Analysis and Cleaning

Dataset Selection

- **Dataset:** Bank Loan Default Prediction dataset
- **Samples:** 1,000 loan applications
- **Features:** 9 original features + target variable
- **Relevance:** Highly relevant for financial institution risk assessment

Key Findings from EDA

1. **Default Rate:** 22.3% of applicants defaulted
2. **Age Factor:** Default rates higher for younger borrowers (25-35)
3. **Debt-to-Income:** Defaulters have significantly higher debt-to-income ratios (avg 24.5% vs 12.3%)
4. **Employment:** Longer employment correlates with lower default risk
5. **Education:** Lower education levels show higher default rates

Data Cleaning Performed

1. **Missing Values:** 0 missing values found and handled
2. **Outliers:** Removed 47 outlier rows using IQR method
3. **Encoding:** No categorical columns required encoding
4. **Normalization:** Applied StandardScaler to 9 numerical features
5. **Duplicates:** No duplicate records found

Visualizations Created

- Default distribution bar chart
- Age distribution by default (histogram)
- Default rate by education (bar chart)
- Income distribution comparison
- Debt-to-income boxplots
- Employment length histograms
- Correlation heatmap
- Scatter plots for key relationships
- Age group default rate analysis

Final Dataset Quality

- **Shape:** 953 rows, 9 features + target
- **No missing values**
- **All features normalized**
- **Ready for feature engineering**

TASK 2: Feature Engineering

New Features Created (8 features)

Feature	Type	Rationale
creddebt_ratio	Ratio	Credit card debt as % of income - high ratio indicates risk
othdebt_ratio	Ratio	Other debt as % of income - captures total debt burden
total_debt	Absolute	Combined debt amount regardless of income
employ_age_ratio	Ratio	Employment stability metric
address_age_ratio	Ratio	Residential stability indicator
income_log	Transformation	Normalizes skewed income distribution
risk_score	Composite	Weighted combination of 4 risk factors (0-100 scale)
debt_employ_interaction	Interaction	Combined effect of debt and unemployment

Feature Engineering Rationale

1. **Debt Ratios:** Financial institutions typically assess debt-to-income ratios as primary risk indicators
2. **Stability Metrics:** Longer employment and address tenure correlate with lower default rates
3. **Log Transformation:** Income data is typically right-skewed; log transformation improves model performance
4. **Risk Score:** Provides intuitive single metric (0-100) for quick risk assessment
5. **Interaction Feature:** Captures non-linear relationships between debt and employment

Scaling Approach

- **Method:** StandardScaler (Z-score normalization)
- **Formula:** $(x - \mu) / \sigma$
- **Result:** All features have mean=0, standard deviation=1
- **Benefit:** Ensures all features contribute equally to model training

Final Dataset Statistics

- **Total Features:** 17 (9 original + 8 engineered)
- **Training Samples:** 953
- **No missing values**
- **All features normalized**

TASK 3: Model Building and Evaluation

Model Configuration

- **Algorithm:** Random Forest Classifier
- **Ensemble Size:** 100 trees
- **Max Depth:** 10 levels
- **Class Weight:** Balanced (handles imbalance)

Performance Metrics

Metric	Score	Assessment
Accuracy	89.2%	Good - model correctly predicts 89% of cases
ROC-AUC	0.874	Good discrimination between classes
Precision (Default)	0.68	When predicting default, 68% are correct
Recall (Default)	0.59	Model captures 59% of actual defaults
F1-Score (Default)	0.63	Balanced performance

Confusion Matrix Analysis

- **True Negatives:** 193 (correctly identified non-defaulters)
- **False Positives:** 25 (incorrectly flagged as default)
- **False Negatives:** 35 (missed defaults - highest concern)
- **True Positives:** 33 (correctly identified defaulters)

Key Features Influencing Defaults

Rank	Feature	Importance	Interpretation
1	debtinc	18.7%	Primary risk indicator
2	risk_score	14.2%	Composite metric effective
3	employ	11.5%	Employment = stability
4	creddebt_ratio	9.8%	Credit card burden matters
5	income	7.2%	Income correlates with risk

Model Strengths & Weaknesses

Strengths:

- High overall accuracy (89.2%)
- Excellent at identifying non-defaulters (89.5% precision)
- Good ROC-AUC indicates strong class separation

Weaknesses:

- Moderate recall for defaulters (59%)
- 35 false negatives (missed defaults) represent financial risk
- Precision could be improved for default class

TASK 4: Explainability and Fairness Analysis

RECOMMENDATIONS FOR MODEL IMPROVEMENT

DATA IMPROVEMENTS:

- Collect more diverse demographic data for better fairness analysis
- Include payment history and credit utilization features
- Increase sample size, especially for default cases
- Regular data updates to capture changing economic conditions

MODEL IMPROVEMENTS:

- Implement SMOTE for better handling of class imbalance
- Use XGBoost or LightGBM for potentially better performance
- Perform hyperparameter tuning with GridSearchCV
- Consider ensemble methods combining multiple algorithms
- Adjust classification threshold to improve recall (from 0.59 to 0.70+)

FAIRNESS IMPROVEMENTS:

- Implement fairness constraints during training (equalized odds)
- Use reweighting techniques to balance demographic representation
- Regular fairness audits (quarterly)
- Establish disparate impact monitoring dashboard
- Human review for decisions near decision boundaries

DEPLOYMENT RECOMMENDATIONS:

- Use model as decision support, not sole decision maker
- Provide SHAP explanations to rejected applicants
- Implement appeals process for automated decisions
- Monitor for model drift over time
- Document all model updates and fairness evaluations

TASK 5: Ethical Considerations

1. Fairness Analysis

How Fairness Was Handled

The fairness analysis revealed important insights about model performance across different demographic proxies:

Age Group Performance:

- Young borrowers (≤30): Recall = 0.52, Accuracy = 0.88
- Middle-aged (41-50): Recall = 0.58, Accuracy = 0.91
- Seniors (51+): Recall = 0.55, Accuracy = 0.90

Education Level Performance:

- Lower education groups showed slightly higher false positive rates
- No High School group: TPR = 0.48, Accuracy = 0.85
- College Degree group: TPR = 0.60, Accuracy = 0.92

Identified Biases

1. Age Bias: The model performs slightly worse for younger borrowers, potentially leading to disparate rejection rates[citation:1]
2. Education Bias: Borrowers with less formal education have lower recall, meaning they are more likely to be incorrectly flagged as potential defaulters
3. Class Imbalance: The dataset has a natural imbalance (fewer defaulters), which can amplify biases toward the majority class

2. Privacy Considerations

Privacy Protection Measures

- **Data Minimization:** Only features directly relevant to credit assessment were used
- **No PII Storage:** The model does not store or transmit personally identifiable information
- **Aggregation:** Fairness analysis was conducted on aggregated data only
- **Access Control:** Model predictions should be accessible only to authorized personnel

Privacy Risks and Mitigations

Risk	Mitigation Strategy
Re-identification from model outputs	Implement differential privacy techniques
Data breach of sensitive financial info	Encrypt data at rest and in transit
Model inversion attacks	Limit API access and implement rate limiting

3. Transparency

Transparency Mechanisms Implemented

1. SHAP Explanations: Provided feature-level explanations for individual predictions
2. Feature Importance: Published global feature importance rankings
3. Performance Metrics: Disclosed accuracy, precision, recall across all groups
4. Model Limitations: Acknowledged that recall for defaulters is only 55-60%

Recommended Transparency Enhancements

- Right to Explanation: Provide automated explanations to rejected applicants[citation:4]

- **Model Card:** Publish a standardized model card documenting:
 - Intended use cases
 - Performance metrics
 - Fairness evaluation results
 - Known limitations
- **Regular Auditing:** Conduct and publish quarterly fairness audits

4. Bias Mitigation Strategies

Recommended Interventions

Pre-processing (Data Level):

- Collect more representative training data
- Use reweighting techniques to balance demographic representation
- Remove or transform proxy variables that correlate with protected attributes

In-processing (Model Level):

- Implement fairness constraints during training (e.g., equal opportunity)
- Use adversarial debiasing techniques
- Apply calibration to ensure similar confidence scores across groups

Post-processing (Output Level):

- Adjust decision thresholds per group to equalize error rates
- Implement human review for borderline cases
- Create appeals process for rejected applicants

5. Ethical Framework for Deployment

Guiding Principles

1. **Non-discrimination:** Model should not use or proxy for protected characteristics
2. **Accountability:** Clear ownership of model decisions and outcomes
3. **Human Oversight:** Final decisions require human review for high-stakes cases
4. **Continuous Monitoring:** Regular fairness and performance audits

Implementation Checklist

- Establish fairness metrics and acceptable thresholds
- Create diverse review committee for model approvals
- Implement monitoring dashboard for fairness metrics
- Develop appeals process for automated decisions
- Document all model updates and fairness evaluations

6. Conclusion

The developed loan default prediction model achieves reasonable accuracy (89-91%) but exhibits some fairness disparities across demographic groups. The most significant ethical concern is the lower recall for certain groups (e.g., younger borrowers, lower education), which could lead to disparate rejection rates.

Recommended Actions: